Paired sign conditions and centered Pell carriers for Diophantine consistency

Research notes for ProveIt

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Abstract

This iteration combines recent ProveIt constructions with a later-stage application of paired sign conditions. If two integers have a known nonnegative sum, one nonnegativity certificate for their product enforces both signs. Applied after a carefully ordered power-recovery argument, this gives a direct quartic in 489 unrestricted integers, compared with the incoming 517-variable predecessor. The variable saving is accompanied by larger coefficient height, not an improvement in every size measure.

A second centering gives a simpler compact endpoint: fifteen unrestricted integers, exact degree 48 and 108 arithmetic operations, compared with degree 56 and 110 operations. Coprime grouping supplies new eleven-integer tradeoffs, while auxiliary Pell growth removes a transport majorant from ten-positive-variable equations. The smallest operation count in that ten-variable family is 89 at degree 288; a degree-254 alternative uses 97 operations. All conclusions concern ordinary mathematical reductions with explicit inherited logical calibration, not new end-to-end kernel certificates or a global optimum.

Principal results

Domain and variables	Degree	Operations	Change or tradeoff
$\mathbb{Z}^{15}$	48	108	Previously 56/110
$\mathbb{Z}^{12}$	500	139	Same polynomial, previously 148 gates
$\mathbb{Z}^{11}$	688	124	Same polynomial, previously 126 gates
$\mathbb{Z}^{11}$	820	119	New intermediate tradeoff
$\mathbb{Z}^{11}$	964	118	New intermediate tradeoff
$\mathbb{Z}^{11}$	1632	112	Same polynomial, previously 114 gates
$\mathbb{Z}^{10}_{>0}$	288	89	Shorter circuit, larger degree
$\mathbb{Z}^{10}_{>0}$	262	94	Previously 266/96
$\mathbb{Z}^{10}_{>0}$	254	97	Incoming predecessor 254/99
$\mathbb{Z}^{489}$	4	—	Direct source, previously 517 variables

Here d/c means degree d and c live binary operations. The dash does not mean zero operations: the direct quartic is compared by its explicit quadratic residuals, expanded terms and coefficient height instead. Additional tradeoffs and exact files are given in the relevant sections. The existing nine-variable positive constructions and shorter circuits with more variables remain separate comparison points. No claim is made that 89 operations or ten variables is a universal minimum.

The direct comparison is particularly important:

Integer variables	Degree	Residuals	Terms	Height
517 (predecessor)	4	310	6774	3904267
493 (paired signs)	4	310	7174	10945213
489 (square anchor)	4	308	6905	11302118

The twelve post-power sign pairs and square anchor extend a contemporaneous upstream six-pair construction; the principle itself is credited, not claimed as new. The fifteen-variable result is an explicit zero-set bijection. Other entries combine existing reductions or optimize evaluators.

Contents

1 Scope and comparison criteria **2**

2 The inspected comparison baseline **3**

3 A different route: a direct quartic in 489 integers **3**
3.1 Two signs need one guard when their sum is known 4
3.2 The signs are recovered after, not before, the powers 4
3.3 An existing square can guard the factorial radius 5
3.4 A self-bounding factorial at any sufficiently large radius 5
3.5 Completeness and the exact polynomial maps 6

4 A fully centered fifteen-integer carrier **6**
4.1 Complete polynomial 6
4.2 Monicity, parity, and exact inverse maps 7
4.3 Exact degree and arithmetic accounting 8

5 A credited compact evaluator for the twelve-integer endpoint **8**

6 Auxiliary-powered coprime Pell fusion **10**
6.1 The complete equations 10
6.2 Root recovery before transport 11
6.3 Why the auxiliary term alone suffices 11
6.4 Soundness and positive completeness 12
6.5 Exact degree, not a syntactic bound 13
6.6 Reproducible artifacts 14

7 Two coprime groups over eleven unrestricted integers **14**

8 Explicit constants and logical dependencies **16**

9 Reproduction and verification boundaries **16**

10 What did not become a result **17**

1 Scope and comparison criteria

We seek a fixed ordinary polynomial P with integer coefficients for which

$$\mathsf{ZFC} \vdash \left(\neg\exists \mathbf{x} \in D^n \; P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

Here the domain D is specified separately for each equation. Positive integer variables are not interchangeable with unrestricted integer variables. Every abbreviation in a displayed defining equation is a polynomial, not an additional existential quantifier. Proof-only rational expressions and radicals do not occur in the literal polynomial circuits.

There is no single size measure that ranks all these constructions. We record the number of variables, exact total degree, and the number of live binary addition, subtraction and multiplication gates. Squaring and multiplication by a constant count as multiplication gates. This is a shared-circuit count, not bit complexity. For the expanded quartics we also record nonzero monomial count and maximum absolute coefficient. Their small coefficients distinguish them from the compact Pell equations, whose few fixed coefficients are very large.

For Pell-sequence notation used below, when $b \geq 2$ and $m \geq 0$ define

$$\chi_b(m) + \psi_b(m)\sqrt{b^2-1} = \left(b + \sqrt{b^2-1}\right)^m.$$

Thus $\chi_b(m)$ and $\psi_b(m)$ are the usual integral Pell coordinates. These proof-only sequences are not operations in the literal polynomials.

The new results are ordinary mathematical reductions, composed with explicitly retained source and syntactic-refutation calibrations. The checks authenticate coefficients, transcriptions, polynomial identities, degree certificates and finite compiler transformations. They do not replace the unbounded proofs. No new end-to-end Lean, Rocq or encoded object-language ZFC certificate is asserted. In particular, the equivalence above is not a proof of either side, and no global minimum or literature-wide priority is claimed.

2 The inspected comparison baseline

The repository snapshot used for comparison is commit `45b399ec51df`, dated 24 October 2026 at 22:28 UTC. It includes the previous coprime-projection packet and seven new peer packets. The source audit authenticates 521 downloaded files against pinned Git blob hashes; this is a selected source snapshot, not an independently rebuilt whole repository.

The canonical displayed equation at that pin has ten positive variables, degree 360 and 98 operations. That display is not the entire research frontier: incoming saturated-Pell constructions include degree 254 at 99 operations, and our preceding packet includes degree 266 at 96 operations. Similarly, the accepted direct quartic has 569 integer variables, while the incoming factorial-carrier packet reaches 517. Comparisons below use the relevant stronger incoming predecessor, not merely the canonical page.

Three useful new ingredients are explicitly credited. The structural compression packet proves that auxiliary Pell growth supplies its own transport bound, and supplies shorter universal norm evaluators. The saturated-Pell packet centers a product of two recovered roots. The powered-saturation factorial packet supplies the direct quartic on which the new sign-guard transformation acts. Our previous coprime-projection packet supplies the compact integer kernels and their complete fixed coefficient bindings.

The newer indexed-tile enumeration shortens a constant recipe, but changes its selector ordering and hence its K coefficient. We do not splice that coefficient into an older tuple. Each compact construction retains one whole calibrated tuple. The archive records the exact predecessor files and checksums so that these distinctions survive later repository edits.

A late comparison at commit `59ee8f212446`, dated 24 October 2026 at 22:32 UTC, already uses six anchored-product sign pairs. It reports an unprojected source with 56 predicates, 1436 coordinates and 1242 rows, saving 18 coordinates and six rows. Accordingly, the elementary paired-sign principle is not claimed as new here. The present direct construction applies twelve pairs at a later power-recovery stage and supplies the projected 493- and 489-variable quartics, the latter with the additional square anchor.

3 A different route: a direct quartic in 489 integers

The direct C222 proof-certificate branch admits a new ordinary polynomial

$$P_{489}(z_1, \ldots, z_{489}) = \sum_{j=1}^{308} R_j(z)^2, \qquad z \in \mathbb{Z}^{489},$$

where the accompanying equation file lists every integral quadratic R_j. Its expanded polynomial has exact degree 4, 6905 nonzero monomials, and coefficient height 11302118. A simpler sibling has 493

integer variables, 310 residuals, 7174 monomials and height 10945213. Both improve the inherited 517-integer direct quartic, not the much-higher-degree eleven-variable frontier.

These are fully materialized polynomials, with no hidden natural variables, variable exponents or unspecified program constants. Their interpretation as consistency equations inherits the existing ordinary C222/ZFC and signed-Pell calibration. Subject to precisely that calibration, the new reductions prove

$$\mathrm{ZFC} \vdash \left(\forall z \in \mathbb{Z}^{489}\ P_{489}(z) \neq 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

No new kernel-checked internal derivation, proof of consistency, global minimality, or historical priority is asserted. The retained source closure includes the finite calculus and its pinned calibration, not just a compiler acting on an unnamed predicate.

Attribution and the different measurement stages. The paired-product principle already appears in the upstream anchored-product construction with six pairs at commit `59ee8f2` (24 October 2026, 22:32 UTC). Its full accepted review reports 56 predicates, 1436 coordinates and 1242 rows in the *unprojected* parent, saving 18 coordinates and six rows. It makes no projected-count claim. Its six pairs use retained scalar anchors before power recovery. Here twelve pairs are restored after power recovery on the later factorial-carrier source; the measured projections are $517 \to 493$ and, with the further F^2 change, 489. We credit the overlapping product-sign principle and claim no priority for it. The authenticated late review is preserved with these artifacts.

3.1 Two signs need one guard when their sum is known

For integers r, s with $r + s \geq 0$,

$$r, s \geq 0 \quad \Longleftrightarrow \quad \exists a, b, c \in \mathbb{Z} : \quad rs = a^2 + b^2 + c(c+1). \tag{PG}$$

Indeed the right-hand side is nonnegative, so a negative r would force $s \leq 0$, contradicting the sum hypothesis; interchange r, s for the other case. Conversely every natural n is $a^2 + b^2 + c(c+1)$: represent $4n + 1$ by three squares, permute its unique odd coordinate to $2c + 1$, and divide the other coordinates by two. This is the usual consequence of Legendre's three-square theorem. The sum hypothesis cannot be omitted: $r = s = -1$ has positive product.

The direct source has twelve digit remainder/slack pairs: two exact-binomial pairs and ten range-binomial pairs. Each has an existing row

$$r + s = q^k - 1,$$

where k is one of the positive indices $t, t+1, y+1, x+1$. Replacing their twenty-four separate three-witness guards by twelve copies of (PG) preserves quadratic residuals. After all monic eliminations, this reduces 517 variables to 493, rather than merely subtracting thirty-six from an unreduced variable count.

3.2 The signs are recovered after, not before, the powers

Keep the factorial ceiling remainder and slack f_r, f_s, and the first exact-digit output slack o_0, separately nonnegative. Their raw rows give

$$D = f_r + f_s + 1 \geq 1, \qquad q = D + o_0 + 1 \geq 2. \tag{B1}$$

All seed-exponent guards and encoded-representative guards are retained. Together with the shared square budget, they establish the signed indexed-Pell hypotheses independently of the twelve removed

digit signs. The request q^t orients the t-indexed pair; thus $V = (\rho+1)^t \geq 1$ has the correct positive value. The unchanged ceiling identity is

$$DF = V + f_s, \tag{B2}$$

which forces $F \geq 1$ and then $V \geq 2$.

The retained guard $B \geq 0$, with $C + F^2 + 1 = Bi_q$ and $C \geq 1$, gives $B > 0$. The signed request B^K gives its positive output N, except possibly for the inverse case $B = 1, N = -1$. In either case the other K-indexed base $N + V + 1$ is at least two. It orients the shared pair, excludes the inverse case, and recovers $N = B^K > 0$ and $q = N(N + V + 1)^K$. Every remaining index has a request at base q or $q + 1$. All nineteen powers are therefore recovered before any replaced sign is used. Now $q^k - 1 \geq 0$, so (PG) restores each original remainder and slack sign. The old carry-free digit, amplified-CRT and Horn-flow proofs apply.

In the reverse direction an old semantic certificate has $r, s \geq 0$, so each product rs receives three witnesses by (PG). This is an existence-preserving map, not an identification of old and new guard coordinates.

3.3 An existing square can guard the factorial radius

The stronger endpoint changes only three source rows:

$$\rho = F^2t^2, \qquad y = F^2 + r_0 + 1, \qquad R = 2 + t + x + y + F^2, \qquad K = 2R. \tag{SQ}$$

The old rows used F in place of F^2 at these three positions. The wire F^2 already exists for the reciprocal product $B = \binom{F^2+t}{t}$. No extra square coordinate is introduced; instead, the independent nonnegativity guard on F is removed.

Before interpreting F, its square gives $\rho \geq 0$, $y \geq 1$ and $R \geq 19$, since $t = t_0 + 16$ and $x, r_0, t_0 \geq 0$. Thus all seed indices remain positive: $t, K, \rho + t, F^2 + t, x + 1, y + 1$, the positive encoded representatives and the positive derived label sum. The new ordered lower-bound certificate establishes these facts directly from the modified polynomial; it does not reuse an assumption $F \geq 0$. Equations (B1)–(B2) and the preceding power recovery now force $F \geq 1$. Only after this step is F used as a natural source parameter.

Once $F \geq 1$, the larger radius satisfies $\rho \geq Ft^2$. The new R dominates the old $2 + t + x + y + F$, so the same local majorant exponent remains valid. When the factorial is recovered, $t \geq 16$ gives $F = t! \geq 16! > 2^{44}$; hence the old bounds

$$L, J < R^E < 2^R < 2^K, \qquad 3\ell(t+1) \leq 2F \leq K$$

continue to validate the source and derived-label range. The literal compiler then removes the factorial guard and one additional monic wire, giving 489 variables.

3.4 A self-bounding factorial at any sufficiently large radius

Let t, F, ρ be positive integers, $\rho \geq Ft^2$, and put

$$V = (\rho+1)^t, \qquad D = \binom{\rho + t}{t}.$$

There are natural f_r, f_s satisfying

$$D = f_r + f_s + 1, \qquad DF = V + f_s \tag{F}$$

if and only if $F = t!$. To prove this write $L = t!$. The exact product identity gives

$$\frac{V}{D} = L \prod_{i=1}^{t} \left(1 - \frac{i-1}{\rho+i}\right), \qquad L\left(1 - \frac{1}{2F}\right) < \frac{V}{D} \le L.$$

The lower bound follows from $\sum(i-1)/(\rho+i) \le t(t-1)/(2\rho) < 1/(2F)$. Rows (F) say $F-1 < V/D \le F$, so $F \le L$. If $F < L$, then $L \ge F+1$ and $L(2F-1) \ge 2F^2$; consequently $L - L/(2F) \ge F$, contradicting the strict lower bound. Thus $F = L$. Conversely, for $F = L$, the ratio lies in $(L-1, L]$; choose $f_s = DL - V$ and $f_r = D - f_s - 1$.

The polynomial does not assume the proposed D is already this binomial. Recover powers first. The common radix satisfies $q > V^2$. Once (B2) has forced $F \ge 1$, one has $\rho \ge t$, and every coefficient through degree t in $(q+1)^{\rho+t}$ is at most $\binom{\rho+t}{t} \le V < q$. Thus the first exact digit is valid and (F) forces $F = t!$. For the second exact binomial use $F^2 + t \le (\rho+1)^2$, giving coefficient bound $V^2 < q$. Only then identify B, derive $B \ge 2$, and validate the remaining ranges. This order avoids a circular interpretation of the product binomial.

3.5 Completeness and the exact polynomial maps

The square-anchored variant is reconstructed from a finite proof, not obtained by keeping the old tuple unchanged. Choose the padded horizon $t \ge 16$ beyond the finite proof-tree length, local parameters and label sums, then set $F = t!$. Construct the positive Horn-flow accumulators with weights $1 + jF$. Choose a multiple y of $t!$ exceeding every accumulator and F^2, so $r_0 = y - F^2 - 1 \ge 0$. The beta CRT gives x and the endpoint quotients. Define ρ, R, K by (SQ), construct the reciprocal CRT representatives at precision $2K$, and then define the powers and radix. The wider-radius lemma supplies the factorial ceilings; the original carry bounds supply all digit witnesses; simultaneous Pell completeness supplies the shared frame. Finally apply (PG) to every product.

The Horn factorization is an exact polynomial identity. Every subsequent coordinate deletion has an integral monic restoration; reversing the recorded list supplies the inverse of projection. The final JSON contains all 308 residuals and all 489 coordinates, and the independent expanded coefficient list verifies degree, support and height. The full proof, deterministic builder, lower-bound certificates, changes to the three radius rows, and both projection/restoration maps accompany this section. Finite regression tests supplement these universal arguments; they do not fabricate a zero or test consistency.

4 A fully centered fifteen-integer carrier

The new endpoint uses fifteen unrestricted integer variables, has exact total degree 48, and admits a 108-gate division-free evaluator (61 multiplications, 25 additions, 22 subtractions). The retained predecessor used degree 56 and 110 gates at the same variable count. This is an ordinary arithmetic equivalence to that predecessor; the inherited native/compiler/refutation theorem, not a new consistency proof or formalized object-language derivation, supplies the connection to Con(ZFC).

4.1 Complete polynomial

The variables are

$$j, C, z, h, r, v, w, T, c, n, f, y, p_1, p_2, p_3 \in \mathbb{Z}.$$

The six numerals $\beta, K_0, M_C, M_F, A_c, B_c$ are the authenticated fixed payloads in the accompanying coefficient directory; M_F is already shifted. All names below are polynomial abbreviations, not extra witnesses:

$$\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \\
X = qW, \quad Y = qv, \quad F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = EYk, \\
I = (2r + 1)c^2, \quad Q = \Delta(f^2 - 1), \quad V = f^2(cT - R) - c, \\
M = 2q^2, \quad U = MH, \quad K_{\mathrm{lo}} = 2X^2 - 3Xc - 2, \\
Z = Un + MK_{\mathrm{lo}} + H(v - q^2), \\
D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2, \\
B_{\mathrm{udget}} = k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \\
P_{\mathrm{os}} = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).
\end{gathered}$$

Define

$$\begin{aligned}
C_{\mathrm{new}} &= (Z + Uk)^2 + 4U(LZ - U), \\
B &= f^2 - \Delta I^2 - 1, \\
C_{\mathrm{aux}} &= V^2 - (Q - 1)y(2V + y) - 1, \\
G &= B_{\mathrm{udget}} - P_{\mathrm{os}}.
\end{aligned}$$

The one equation is

$$\boxed{P = C_{\mathrm{new}}^2 + B^2 + C_{\mathrm{aux}}^2 + G^2 = 0.}$$

In particular, no inequality, rational-domain variable, division, or free coefficient port occurs in the literal final polynomial.

4.2 Monicity, parity, and exact inverse maps

The pinned $\beta = 2^{1064921} - 1$ is odd and $1 \pmod 3$. Hence q is nonzero, even, and $2 \pmod 3$. Also H is nonzero and odd, and $H \equiv 3 \pmod q$. Thus $\gcd(H, 2q^2) = 1$ and $U \neq 0$ on every signed assignment. At a zero of P, put $t = Z/U \in \mathbb{Q}$. The new carrier is equivalently

$$t^2 + 2(2L + k)t + k^2 - 4 = 0.$$

Monicity gives $t \in \mathbb{Z}$ before any Pell or sign argument. Reducing $U \mid Z$ modulo the coprime factors M, H yields

$$v_0 = \frac{v - q^2}{M} \in \mathbb{Z}, \qquad g = \frac{K_{\mathrm{lo}}}{H} \in \mathbb{Z}, \qquad t = n + g + v_0.$$

Reduction of the monic equation modulo 2 shows $t + k$ is even. Therefore $s = (t + k)/2$ is an integer. Equivalently, $(t + 2L + k)^2 = 4[1 + L(L + k)]$, and its integer square root is even.

The predecessor differs only in its carrier and its n coordinate:

$$C_{\mathrm{old}} = Z_{\mathrm{old}}^2 + (2Z_{\mathrm{old}} - Uk)UL - U^2.$$

The universal polynomial identity

$$C_{\mathrm{new}}(2Z_{\mathrm{old}} - Uk) = 4C_{\mathrm{old}}(Z_{\mathrm{old}})$$

proves both directions. From a new zero keep every other input and set

$$n_{\text{old}} = \frac{n_{\text{new}} - g - v_0 + k}{2} = s - g - v_0 \in \mathbb{Z}.$$

Then $Z_{\text{old}}/U = s$ and $C_{\text{old}} = 0$. Conversely the old carrier is monic in Z_{old}/U; it therefore licenses the same CRT offsets at every old zero. Set

$$n_{\text{new}} = 2n_{\text{old}} + g + v_0 - k.$$

Then $Z_{\text{new}} = 2Z_{\text{old}} - Uk$ and $C_{\text{new}} = 0$. The other three residuals do not involve n and remain zero in both directions. These maps are mutually inverse on the zero sets. Divisions in the proof maps are licensed at zeros; none is hidden in the polynomial.

4.3 Exact degree and arithmetic accounting

The relevant degrees are

$$\deg U = 6, \quad \deg Z = 7, \quad \deg L = 11, \quad \deg k = 5.$$

The unique degree-24 term of C_{new} is $4ULZ$. The four residual degrees are $24, 14, 18, 16$, so $\deg P \le 48$. Writing leading homogeneous parts with subscript 0 gives

$$(C_{\text{new}})_{24} = 4U_0^2 L_0 n, \qquad P_{48} = 16U_0^4 L_0^2 n^2,$$

where

$$q_0 = 6\beta j, \qquad X_0 = 12q_0 w, \qquad Y_0 = q_0 v,$$
$$U_0 = 8q_0^2 X_0 Y_0, \qquad L_0 = X_0 Y_0^2 (R_0 + hX_0 Y_0).$$

Because R is independent of h, the actual coefficient is

$$[j^{26} w^8 v^{10} h^2 n^2] P = 2^{58} 3^{34} \beta^{26} > 0.$$

Its exponents sum to 48; the other squared residuals have smaller degree. This proves exact degree after the huge constants are fixed.

The evaluator has 108 live binary gates and all fifteen inputs are live. Squares and scalar multiplications are charged. Exact register-cut identities, source/payload hashes, leading forms, liveness, counts, and deliberately bad mutations are checked independently of the builder, under normal and optimized Python. These finite certificates do not replace the unbounded inherited compiler/refutation theorem or assert global minimality.

5 A credited compact evaluator for the twelve-integer endpoint

The retained twelve-integer degree-500 carrier has a pointwise-identical 139-gate evaluator, improving its prior 148 gates without changing any zero. The count is 84 multiplications, 32 additions, and 23 subtractions. This is an evaluator adaptation, not a new semantic twelve-variable reduction.

Credit for the 28-gate universal four-root norm schedule belongs to the existing repository proof at commit `84e16e430926`, `candidates/ten-positive-zero-offset/source/ORIGINAL_PROOF.md`. The received structural-compression addendum explains the schedule; both texts and the upstream license are retained. The new adaptation combines that schedule with the already verified lower-degree centered carrier.

The twelve coordinates are

$$j, C, z, h, r, v, w, T, c, n, p_2, p_3 \in \mathbb{Z}.$$

Use the preceding section's definitions of $J, q_m, q, W, X, Y, F, R, E, a, H,\ \Delta, k, \eta, L, I, M, U, K_{\mathrm{lo}}, Z, D_{\mathrm{err}}$ and B_{udget}, with the same six fixed numerals. Replace its f-dependent definitions by

$$D_f = 1 + \Delta I^2, \qquad Q = \Delta(D_f - 1), \qquad V = D_f(cT - R) - c.$$

Then put

$$\begin{aligned} D_\tau &= 1 + L(L + k), & D_a &= (Q - 1)(QV^2 - 1), \\ G_0 &= B_{\mathrm{udget}} - 1 - p_2^2 - p_3(p_3 + 1), \\ Z_{\mathrm{total}} &= Z + U\big(2L + k + 6(a + 2)I + 9(2Q - 1)V\big). \end{aligned}$$

These definitions reproduce the complete prefix in `lowdegree/twelve/specification.json`.

For the norm evaluator only, introduce five local arguments

$$\widehat{z} = Z_{\mathrm{total}}^2, \quad \widehat{a} = 4U^2 D_\tau, \quad \widehat{b} = 36U^2 D_f, \quad \widehat{c} = 324U^2 D_a, \quad \widehat{d} = 729U^2 G_0.$$

The hats distinguish these temporary arguments from the original coordinates and prefix abbreviations. Set

$$\begin{aligned} \sigma_1 &= \widehat{a} + \widehat{b} + \widehat{c}, & \sigma_2 &= \widehat{a}\widehat{b} + (\widehat{a} + \widehat{b})\widehat{c}, & \sigma_3 &= \widehat{a}\widehat{b}\widehat{c}, \\ U_c &= \widehat{z} + \widehat{d}, & P_o &= U_c - \sigma_1, \\ P_e &= P_o^2 + 4(\widehat{z}\widehat{d} - \sigma_2), \\ G_e &= P_e^2 + 16(\widehat{z}\widehat{d}P_o^2 - 4\sigma_3 U_c), \\ G_o &= P_e P_o - 16\sigma_3. \end{aligned}$$

The compact output is

$$\boxed{N_c = G_e^2 - 64\widehat{z}\widehat{d}G_o^2.}$$

Sharing $\widehat{a} + \widehat{b}$, $\widehat{a}\widehat{b}$, P_o^2 and $\widehat{z}\widehat{d}$ gives exactly 28 binary gates in the tail; the unchanged prefix needs 111.

For the exact proof, the three-root norm is

$$N_3(\xi) = \big[(\xi - \sigma_1)^2 - 4\sigma_2\big]^2 - 64\sigma_3\xi.$$

Put $\xi = U_c \pm \varrho$, where $\varrho^2 = 4\widehat{z}\widehat{d}$. Direct substitution gives $N_3(U_c \pm \varrho) = G_e \pm 4\varrho G_o$. Their product is N_c, proving that the compact formula is the full sixteen-sign norm, including every singular specialization. It equals the predecessor's norm as a universal integer polynomial.

The identity map on all twelve coordinates is therefore a two-way map between the new and predecessor zero sets. All inherited unbounded squareclass, CRT, guard, and native/compiler/refutation arguments remain unchanged. Exact degree 500 and the genuine fixed-coefficient witness

$$[j^{228}w^{76}v^{84}h^8z^8r^{32}c^{64}]N_c = 2^{488}3^{332}\beta^{228} > 0$$

transfer by pointwise equality. An independent checker compares the exact live prefixes up to associative/commutative rewrites and all 495 universal tail monomials, verifies live-node counts, and rejects a changed opcode at every one of the 139 live gates. No claim of new consistency proof, global optimum, or new end-to-end proof-assistant certification is made.

6 Auxiliary-powered coprime Pell fusion

The auxiliary Pell solution supplies a transport bound large enough to remove a separate majorant from the coprime-fused construction. Combining this fact with upper or sharp centering gives the following six explicit equations. All counts charge every live binary addition, subtraction and multiplication, including multiplication by fixed integers.

Chart and center	Positive unknowns	Operations	Exact degree
$X = qw$, no center	10	89	288
$X = qw$, upper center	10	91	283
$X = qw$, sharp center	10	92	278
$X = q + x$, no center	10	94	262
$X = q + x$, upper center	10	96	258
$X = q + x$, sharp center	10	97	254

The incoming saturated-Pell packet already has fusion and centering, and the incoming structural-compression packet already has the auxiliary index bound. The improvement here is their compatible combination, together with the direct-chart contraction $q^4(q-1) = q^3[q(q-1)]$. In particular the centered direct circuits reduce 98 to 96 operations at degree 258, and 99 to 97 at degree 254. The ordinary centered circuits reduce 92 to 91 operations at degree 283, and 93 to 92 at degree 278. These are local improvements, not claims of a global minimum or a new nine-variable construction.

6.1 The complete equations

The ten unknowns are $J, F, \alpha, i, T, x, Y, n, \eta, \zeta > 0$ on the direct chart. On the ordinary chart replace x by $w > 0$. The fixed coefficients are exactly the complete original compact44 tuple in `positive/coefficients/`; its K_0 is not the different incoming BFS K_0. Write

$$
\begin{aligned}
q &= \beta J + 1, & X &= q + x \quad \text{or} \quad qw, & \mu &= q^2 + 1,\\
A &= \mu Y(X+1) + 1, & \Delta &= A^2 - 1, & H &= 4A - 5,\\
k &= \eta + \zeta, & c &= kY + \eta, & v_0 &= X\mu^2 Y^2,\\
D &= v_0(v_0+1)k^2 + 1, & E &= ic^2, & F_s &= 1 + \Delta DE^2,\\
Q &= \Delta^2 DE^2, & \delta &= Q - 1, & C &= q - F - \alpha,\\
G &= (q-1)(q-F) + \alpha, & R &= G(q^2-1) + (M_C + qM_F)J, & e &= k - R - 1,\\
b &= c(T+\alpha) - R, & V &= F_s b - c, & D_y &= QV^2 - 1,\\
K_+ &= X(3c - 2X) + 2.
\end{aligned}
$$

Choose one of the three polynomial centers

$$S = 0, \qquad S = (A+1)DE, \qquad S = ADE.$$

On the direct chart set $B = (qK_0 + X)C - qF$, $P = q^4(q-1)$ and

$$Z = P\mu^2\{H(n + V + S - bB) + K_+\} + H\{\mu^2[q(q-1)Y - q^3B - (q-1)X] - Pe\}.$$

On the ordinary chart set $B_t = (K_0 + w)C - F$, $P = q^3(q-1)$ and

$$Z = P\mu^2\{H(n + V + S - bB_t) + K_+\} + H\{\mu^2[(q-1)Y - q^3B_t] - Pe\}.$$

For either chart put

$$M = P\mu^2 H, \qquad s = \delta Z^2, \qquad u = \delta M^2 D F_s, \qquad v = M^2 D_y, \qquad \boxed{N = (s - v - u)^2 - 4uv = 0.}$$

Every displayed abbreviation is a polynomial, not an extra quantified unknown. The six JSON files give the full literal arithmetic circuits. The complete source proof and coefficient bindings are retained locally.

6.2 Root recovery before transport

On every positive input, $D > 1$, $E \geq c^2$, $F_s > c + 2$ and $\delta > 0$. The integer b is either zero, giving $V = -c$, or satisfies $|V| \geq F_s - c > 1$. Thus $a = DF_s$ and $d_y = D_y/\delta$ are positive. They cannot be equal: $\delta a = D_y$ is impossible modulo D, since $D_y \equiv -1 \pmod{D}$.

The four-factor identity is

$$(z^2 - a - d_y)^2 - 4ad_y = \prod_{\epsilon,\nu \in \{-1,1\}} (z - \epsilon\sqrt{a} - \nu\sqrt{d_y}).$$

Applied with $z = Z/M$, it gives a rational signed sum at a zero. This sum is nonzero because the radicands differ. Squaring and solving linearly shows that both roots are rational. Since a is an integer and $\gcd(D, F_s) = 1$, the factors are squares:

$$D = \tau^2, \qquad F_s = f^2, \qquad I = \tau E, \qquad f^2 - \Delta I^2 = 1, \quad c^2 \mid I, \quad Q = (\Delta I)^2.$$

The adjacent-Pell descent makes $y = \sqrt{d_y}$ integral. Indeed, write $y = a_0/m$ in lowest terms with $m > 0$. Then $m^2 \mid Q - 1$, and the norm-preserving transformation

$$v' = (2Q - 1)v - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2Qv$$

keeps v' integral and the denominator m, while $0 < |v'| < v$ for $v > 1$. Descent from $v = |V|$ reaches $v = 1, y = 1$. Therefore

$$Z/M = \epsilon\tau f + \nu y. \tag{P1}$$

The retained all-index folding lemma now gives $c = \psi_A(p)$ with p odd. The first Pell equation gives $k = 2\psi_{2v_0+1}(n_0)$. If $n_0 = 1$, then $\eta = \zeta = 1$ and $c = 2Y + 1$ lies strictly between $\psi_A(1)$ and $\psi_A(2)$, a contradiction. Hence

$$k \geq 8v_0 + 4, \quad c > kY, \quad c \geq \psi_A(3) = 4\Delta + 3.$$

The unchanged pre-transport mask estimate is $R < c\alpha/4$. It gives

$$b > c(T + 3\alpha/4) > c, \qquad V > 1, \qquad K_+ > 0. \tag{P2}$$

These steps do not assume transport nonnegativity, the main-index equality, evenness of q, or divisibility of e.

6.3 Why the auxiliary term alone suffices

Set $S_0 = \Delta I$, so $Q = S_0^2$. Pell classification of the auxiliary norm gives $S_0 V = \chi_{S_0}(\nu_0)$. Reduction modulo S_0 shows that ν_0 is odd. By (P2),

$$V = F_s b - c > (F_s - 1)c = \Delta I^2 c > 4Q.$$

The indices 1 and 3 give $V = 1$ and $V = 4Q - 3$, respectively. Thus $\nu_0 \geq 5$, and

$$V \geq 16Q^2 - 20Q + 5 > 8Q^2, \qquad b = (V + c)/F_s > 8Q. \tag{P3}$$

Here $F_s < Q$. Also $Q - 1 > 2F_s$, so

$$0 < y - V = \frac{V^2 - 1}{(Q-1)(y+V)} < \frac{b}{4}, \qquad \frac{e}{\mu^2} < \frac{b}{4}. \tag{P4}$$

The second inequality follows from $b - e = c(T + \alpha) - k + 1 > 0$ and $\mu^2 > 4$; it is an upper bound even when $e < 0$. Finally

$$\tau f < ADE < Q, \qquad X < D < Q, \qquad \tau f + X/q^4 < 2Q < b/4. \tag{P5}$$

On the direct chart the exact center is

$$\frac{Z}{M} = n + V + S + \frac{K_+}{H} + \frac{Y}{q^3} - \frac{WB}{q(q-1)} - \frac{X}{q^4} - \frac{e}{\mu^2}, \qquad W = 1 + q(q-1)b.$$

If $B \le -1$, then $WB/[q(q-1)] < -b$. Solve this relation and (P1) for n. Since $S \ge 0$, (P4)–(P5) give

$$n < b/4 + b/4 + b/4 - b = -b/4 < 0,$$

a contradiction. Thus $B \ge 0$. The ordinary chart has $W = 1 + (q-1)b$ and omits X/q^4; the same proof gives $B_t \ge 0$.

6.4 Soundness and positive completeness

For the direct chart, the complete old fused polynomial is recovered by the universal substitution

$$n_{\text{old}} = n + S + HDE\,B. \tag{P6}$$

The old transport majorant was $HDE + b$, so (P6) makes the entire carrier and norm identical at every integer assignment. At a new positive zero, $B \ge 0$, hence $n_{\text{old}} > 0$. The retained old soundness theorem therefore gives a finite syntactic refutation.

For the ordinary chart, the now-licensed inherited proof gives successively $C > 0$, $3q < R < q^4 - 2$, the main index $p = R$, even q, $\mu^2 \mid e$, and the first index $(R+1)/2$. The exact ratio estimate and whole-product projection give

$$H \mid q^3(q-1)(X - 2^R)(2^R X + 1),$$

with the complete product smaller than H in absolute value; hence $X = 2^R$. No potentially nonunit factor is canceled modulo H. The residual carrier fractions give $q^3 \mid Y$ and $q - 1 \mid B_t$, while fixed-minus parity gives $R \equiv 3 \pmod 4$. Fresh unscaled source reconstruction uses $A_{\text{source}} = Y(X+1) + 2$. The same coherent native calibration gives a finite refutation.

Conversely, start with the retained canonical source witnesses from a finite refutation. They have $X = 2^R$, $R \equiv 3 \pmod 4$, $q^3 \mid Y$, $q^4 \mid X$, $D = \tau^2$, and positive first/main coordinates. Choose a positive Pell matrix period modulo τc^2, and set

$$I = \psi_A(m), \quad f = \chi_A(m), \quad i = I/(\tau c^2) > 0.$$

The finite order of an invertible matrix over this finite residue ring guarantees such a period. The new literal kernels are then exactly $F_s = f^2$ and $Q = (\Delta I)^2$.

Use the explicit auxiliary-completeness witness

$$V_* = \frac{\chi_{\Delta I}(R)}{\Delta I}, \qquad y = \psi_{\Delta I}(R), \qquad T_* = \frac{V_* + c + Rf^2}{cf^2}, \qquad T = T_* - \alpha.$$

The retained lemma proves integrality. Its elementary growth estimate gives $V_* \geq 9Q^2$ and $T_* > 9\Delta I > \alpha$. Hence $T > 0$, and

$$h_y := y - V_* > \frac{V_*}{4Q} \geq \frac{9Q}{4}. \tag{P7}$$

Write $\kappa = K_+/H$, $s_Y = Y/q^3$ and $\rho = e/\mu^2$; these are canonical integers, with $0 < \kappa < c < I$, $0 < s_Y < c < I$ and $\rho \geq 0$. The canonical transport quotient t is nonnegative. All three choices of center obey $S - \tau f < 2DE < 2I^2$. Define

$$n = Wt + \rho + \tau f - S + h_y - \kappa - s_Y + \begin{cases} X/q^4, & \text{direct chart,} \\ 0, & \text{ordinary chart.} \end{cases}$$

The negative loss is less than $2I^2 + 2I < 4I^2$, while (P7) exceeds $4I^2$. Thus $n > 0$, even when $t = 0$, and the center equals $\tau f + y$. This proves the converse with genuinely positive coordinates, not a presumed positive inverse of (P6).

The equivalence to the absence of finite ZFC refutations inherits the exact retained source calibration. This ordinary mathematical argument does not assert that a new Lean endpoint, an internal encoded ZFC certificate, or a proof of consistency has been produced.

6.5 Exact degree, not a syntactic bound

Set $q_0 = \beta J$, $k_0 = \eta + \zeta$ and $X_0 = q_0 + x$. The direct leading homogeneous forms are

$$\begin{aligned} [N_0]_{262} &= -2^{10} q_0^{126} X_0^{30} Y^{62} i^{10} k_0^{32} (q_0 - F)^2, \\ [N_+]_{258} &= 2^{11} q_0^{124} X_0^{29} Y^{61} i^{10} k_0^{32} (q_0 - F)^2, \\ [N_A]_{254} &= 2^{10} q_0^{122} X_0^{28} Y^{60} i^{10} k_0^{32} (q_0 - F)^2. \end{aligned}$$

The ordinary leading forms are

$$\begin{aligned} [N_0^{\mathrm{ord}}]_{288} &= -2^{10} q_0^{152} w^{30} Y^{62} i^{10} k_0^{32} (q_0 - F)^2, \\ [N_+^{\mathrm{ord}}]_{283} &= 2^{11} q_0^{149} w^{29} Y^{61} i^{10} k_0^{32} (q_0 - F)^2, \\ [N_A^{\mathrm{ord}}]_{278} &= 2^{10} q_0^{146} w^{28} Y^{60} i^{10} k_0^{32} (q_0 - F)^2. \end{aligned}$$

They are nonzero for the actual coefficients because $\beta > 0$.

For an exact global certificate, scale every unknown by a formal t and work in the Laurent-series field at $t = \infty$. Choose roots $g^2 = DF_s$ near ADE and $y^2 = D_y/\delta$ near V. The identities

$$(ADE - g)(ADE + g) = D^2E^2 - D, \qquad (y - V)(y + V) = (V^2 - 1)/(Q - 1)$$

control all cancellations. On the direct chart the degrees of (δ, M, V, g) are $(42, 13, 38, 25)$; the defects $ADE - g$ and $(A + 1)DE - g$ have degrees 17 and 21. On the ordinary chart these are $(48, 13, 42, 28)$ and $18, 23$. The rational carrier remainder after $V + S$ has degree at most 6, while $y - V$ has degree -4 or -6. The exact identity

$$N = \delta^2 M^4 \prod_{\epsilon,\nu \in \{-1,1\}} (Z/M + \epsilon y + \nu g)$$

therefore gives all six degrees and highest forms above. For example, direct sharp centering has degree

$$2(42) + 4(13) + 2(38) + 25 + 17 = 254.$$

This proves the global upper bounds and attainment; exact univariate replay is an additional transcription check, not the degree proof itself.

6.6 Reproducible artifacts

The full proof, explicit canonical witnesses, coefficient payloads, all six live gate DAGs, and the exact builder are in `positive/`. Run `build.py --check`, `check.py` and `check_ordinary.py`, both normally and with Python's `-O` flag. The recorded receipts pass coefficient hashes and bit lengths, whole graph inventories, liveness, universal carrier identities, exact integer ray expansions, actual-payload modular leading forms, independent formula evaluations and finite Pell-period controls. The unbounded proof is independent of those finite tests. All copied upstream material and its attribution remain in `positive/source/`.

7 Two coprime groups over eleven unrestricted integers

The new integer frontier is

$$(\deg, \mathrm{ops}) \in \{(688, 124), (728, 120), (820, 119), (964, 118), (1632, 112)\}.$$

Every equation uses the same eleven unrestricted integer coordinates and six fully instantiated numeral coefficients. The degree-820 and 964 points are new mathematical reductions; the other three evaluate the previous polynomials with two fewer binary operations. No ten-integer endpoint, degree below 688, global optimum, proof of consistency, or new formal kernel certificate is claimed.

The eleven independent coordinates, all in $\mathbb{Z}$, are

$$j, C, z, h, r, v, w, T, c, n, s.$$

The six fixed integers are $\beta, K_0, M_C, M_F, A_F, A_C$, with

$$\beta = 2^{1064921} - 1, \qquad A_F = 2^{1113}, \qquad A_C = 2^{666104}.$$

Here K_0, M_C, M_F are the actual authenticated numeral payloads in `integer/coefficients/BINDING.json`, not parameters; M_F is already shifted. Their complete finite reconstruction (including the machine and tile data) is retained in `integer/coefficients/recipe/`. The checker rebuilds the full tuple and compares every payload byte. Define

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4,\\
X &= qW, & Y &= (2v + 1)q^3, & E &= XY,\\
a &= E + Y, & A &= a + 2, & H &= 4a + 3,\\
\Delta &= a^2 + H = A^2 - 1, & F &= (K_0 + W)C - z(q - 1).
\end{aligned}$$

Further put

$$\begin{aligned}
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J,\\
k &= R + 1 + hE, \qquad \eta = c - Yk,\\
D_{\mathrm{err}} &= (A_F F - q)^2 + (A_C C - q)^2,\\
\mathrm{Budget} &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2,\\
D &= 3\mathrm{Budget} - 4, \qquad \ell = XY^2, \qquad L = \ell k,\\
D_\tau &= 1 + L(L + k) = 1 + \ell(\ell + 1)k^2.
\end{aligned}$$

Every displayed symbol other than the eleven coordinates is a fixed numeral or polynomial abbreviation, not an additional unknown. The positivity group is

$$D_g = 1 + 4DD_\tau s^2, \qquad P = D_\tau D_g = D_\tau + D(2sD_\tau)^2.$$

For the new pair put

$$D_m = 1 + \Delta c^2, \quad I = 2src^2AD_m, \quad D_f = 1 + \Delta I^2, \quad Q = \Delta^2 I^2, \quad \delta = Q - 1,$$

$$V = D_f(c - R + 4IT) - c, \quad C_y = QV^2 - 1, \quad D_a = \delta C_y,$$

$$K = X(3c - 2X) + 2, \quad S = IH^5, \quad z_c = HT + K, \quad A_c = (z_c^2 + 3H^2)(z_c^3 - 19H^3), \quad Z = Sn - A_c.$$

The new literal integer polynomials are

$$N_{\text{int}} = (Z^2 - S^2 D_f D_a - 4S^2 P)^2 - 16S^4 D_f D_a P,$$
$$N_{\text{rat}} = (\delta Z^2 + 4\delta S^2 P - S^2 D_f C_y)^2 - 16\delta^2 S^2 P Z^2.$$

The latter is denominator-cleared, not a rational operation.

Soundness. At $I = 0$, one has $S = Q = 0$ and $\delta = -1$; thus the rational clearing factor δ^2 equals one and both polynomials equal $A_c^4 \neq 0$, since $(t^2 + 3)(t^3 - 19)$ has no rational root and $H \neq 0$. Thus $I, s, r, c \neq 0$. The established bootstrap gives $D_f, D_a > 0$. A vanishing conjugate has no sole nonzero imaginary term, hence $D_g > 0$, which forces Budget ≥ 2 and the centered signed packing bounds. For N_{int}, a signed rational radical sum is $\sqrt{D_f D_a} + 2\sqrt{P}$; for N_{rat}, multiplication by δ gives $\sqrt{D_f D_a} + 2\delta\sqrt{P}$, with the conjugate signs understood. All underlying radicands and δ are odd. The distinct coefficient 2-adic valuations force both products to be squares. Now $\gcd(D_\tau, D_g) = 1$, while modulo D_f,

$$Q \equiv -\Delta, \quad \delta \equiv -A^2, \quad V \equiv -c, \quad C_y \equiv -D_m, \quad D_a \equiv A^2 D_m.$$

Since $AD_m \mid I$, one has $\gcd(D_f, D_a) = 1$. Each of D_f, D_a, D_τ, D_g is therefore a square. The retained adjacent-coefficient Pell descent makes $y = \sqrt{D_a}/\delta$ integral. Consequently $S \mid A_c$. Monic prime-adic comparison yields $H \mid K$, and $K = HXc - 2[X(X + 2ac) - 1]$ restores the same main projection. The retained signed folding, nonodd strong-rank, whole-product, centered orientation, and odd-witness regeneration arguments apply with exactly their original hypotheses, yielding an old calibrated refutation witness. The full statements and every hypothesis check are in `integer/PROOF.md`.

Completeness. Fix canonical outer data first. Since $DD_\tau > 0$ and $DD_\tau \equiv 2 \pmod 3$, Pell existence provides $s, g > 0$ with $g^2 = 1 + 4DD_\tau s^2$. Apply the retained full-residue interpolation theorem with $N = 2sAD_m$. It fixes a single strong pair I, f satisfying $8sAD_m c^2 \mid I$, with every auxiliary T class modulo I still available. Set $r = I/(2sAD_m c^2)$. Since $H \mid K$, write $\kappa = K/H$. Intersectivity chooses a root of $(T + \kappa)^2 + 3$ times $(T + \kappa)^3 - 19$ modulo I, and interpolation realizes that class without changing I, f. Thus $A_c/S \in \mathbb{Z}$. Finally choose $n = A_c/S + f\delta y + 2\tau g$ or $n = A_c/S + fy + 2\tau g$, respectively. This proves both existence directions with the same explicitly inherited ordinary native/compiler/refutation calibration.

Exact costs. The degrees $\deg I = 24$, $\deg S = 54$, $\deg D_f = 60$, $\deg Q = 72$, $\deg V = 85$, $\deg C_y = 242$, $\deg D_a = 314$ give unique maximal terms $(S^2 D_f D_a)^2$ and $(S^2 D_f C_y)^2$. Their exact degrees are 964 and 820, at 118 and 119 operations. For the degree-688 and degree-728 endpoints, instead put $I = 2src^2$ and recompute $D_f, Q, \delta, V, C_y, D_a, S$ from their displayed formulas. Set

$$u = S^2 D_f, \quad b = 4S^2 D_a, \quad e = 16S^2 P, \qquad Z = S(n + \sigma) - A_c,$$

where $\sigma = 0$ for degree 728 and $\sigma = (2Q - 1)V$ for degree 688. The equation is the balanced octic

$$[(Z^2 + b - u - e)^2 - 4(Z^2 b + ue)]^2 - 64(Z^2 b)(ue) = 0.$$

The exact identity with the prior eight-conjugate polynomial and the direct expression for P save two gates, giving 728/120 and 688/124. For the degree-1632 endpoint, put $I = 2src^2PD_mA$, recompute the same downstream expressions, set $Z = Sn - A_c$, and use

$$\delta Z^2 - S^2 D_f P C_y = 0.$$

The multiplier makes D_f, D_τ, D_g, D_a pairwise coprime. Thus a square product recovers their individual squares, and the same signed soundness and fixed-pair interpolation proof applies. Sharing P in both the multiplier and output gives 1632/112. Independent standard-library audits bind every live register in a free polynomial ring, multiply all four/eight conjugates, rebuild the actual numerals, and certify exact leading monomials and centered cancellations; they pass unchanged under `python -O`.

8 Explicit constants and logical dependencies

The compact integer equations retain the six original-label source34 coefficients: $\beta = 2^{1064921} - 1$, K_0, M_C, the already shifted M_F, and the two cone coefficients 2^{1113} and 2^{666104}. The positive equations retain the complete original compact44 tuple, with $\beta = 2^{590069} - 1$ and its own K_0, M_C, M_F. These are two different complete tuples, not interchangeable collections of ports. Their actual binary payloads, hashes and finite reconstruction recipes are supplied. A fixed named numeral is not an adjustable parameter.

The direct quartic instead supplies every small integer coefficient in its residual and expanded-polynomial files. Its source closure contains the finite C222 calculus, fixed source descriptions, ordinary calibration proofs and pinned formal-system sources. The new local proof changes sign guards and the factorial radius, then uses exact Horn factorization and monic projection. It does not infer source semantics merely from a successful compiler run.

More precisely, the direct review checks the complete factorial, derived-label and hybrid/Horn-flow proofs. The 489-variable result establishes the local source transformation and literal polynomial correctness relative to retained theorem interfaces: the literal square-shell constructor/schema-to-C222 calibration and its raw-builder/majorant binding, together with the simultaneous indexed-Pell/calculus bridge. These are substantive inherited ordinary-proof dependencies. Neither this packet nor its independent local review freshly proves the entire transitive calibration chain; the limitation is not merely absence of a kernel certificate.

For every construction the logical chain is explicit:

$$\exists \text{ polynomial zero} \iff \exists \text{ accepted finite source certificate} \iff \exists \text{ finite syntactic ZFC refutation.}$$

The refutation calibration is inherited ordinary mathematics, not a fresh transitive reconstruction here or an internal proof-assistant derivation. Negation gives the claimed consistency equivalence in ordinary mathematics formalizable in ZFC. A reader who does not accept an inherited calibration still has the separately stated new zero-set reductions and their explicit dependencies; a checksum is not offered as a substitute for that theorem.

9 Reproduction and verification boundaries

The archive includes the TeX source, compiled PDF, literal equations, coefficient payloads, mathematical proofs, generators, independent audits and the immutable preceding packet needed by relative source references. The root reproduction program works in isolated temporary copies rather than changing the delivered originals. It runs both ordinary and optimized Python, compares deterministic rebuilt files and exercises corruption controls. The replay receipt records exactly which commands were run and which files were rebuilt byte-for-byte.

There are three distinct levels of evidence. First, the mathematical proofs establish unbounded implications, including the order of sign, index and divisibility recovery. Second, independent symbolic and compiler checks establish identities and the literal transcription of the claimed equations. Third, finite numerical and mutation tests provide regression and error-detection evidence. The third level cannot replace the first. The final report is compiled and every page visually inspected separately from these algebraic checks.

The direct guard transformation is reviewed independently of its producer. The fifteen-variable equivalence has an independent complete register, degree, provenance and inverse-map audit. The positive family has an independent six-chart transcription and proof review, including the correct negative-transport chronology and fresh positive completeness. The eleven-variable pair is reviewed against its fixed-pair interpolation theorem, not an unsupported assertion that arbitrary Pell witnesses can be synchronized.

10 What did not become a result

No new nine-positive-variable or ten-unrestricted-integer endpoint was obtained. A lower-degree carrier for the eleven-variable construction would be valuable, but replacing its intersective quintic by a quadratic is not justified merely by finding roots modulo some test moduli. The strong ordinate must retain prescribed divisors, and its zero branch must be excluded. Similarly, forcing a Pell guard ordinate to be odd does not follow from taking powers of a unit. These are unresolved directions, not proved universal barriers.

The completed gains are therefore specific: a twelve-pair sign compression with a checked late-stage bootstrap, a short zero-set bijection lowering one compact degree, two new coprime-group tradeoffs, and compatible transport and evaluator simplifications. They do not establish a unique "simplest" equation across incomparable domains, degrees, heights and circuit costs.

References

[1] V. Reshetnikov, *ProveIt, Diophantine consistency research*, snapshot `45b399ec51df`, 24 October 2026. Pinned repository source.

[2] *Coprime Pell products and rational projection for Diophantine consistency*, preceding research packet, 24 October 2026. The exact released packet is retained under `prior/`; its archive SHA-256 is recorded in the root dependency record.

[3] *ConZFC Saturated Pell*, incoming packet, 24 October 2026. Product-root saturation, upper and sharp centers, and canonical positive completeness; cited source sections are retained in `positive/source/`.

[4] *ConZFC Structural Compression*, incoming packet, 24 October 2026. Auxiliary-index transport bound and compact norm evaluators; retained with exact source attribution.

[5] *ConZFC Powered Saturation Factorial Carriers Round 14*, incoming packet, 24 October 2026. Fixed direct source, factorial carrier, derived label ranges and the 517-integer predecessor; required proof closure is retained in `alternative/inherited_proofs/`.

[6] *Anchored-product sign-pair comparison*, late repository snapshot `59ee8f212446`, 24 October 2026, 22:32 UTC. Pinned repository comparison.

[7] *Source and calibration ledger*, accompanying artifact. Each component's `SOURCE_PINS.json`, coefficient binding and retained proof files specify the imported theorem interfaces and their exact provenance. These local references are part of the release, not an appeal to a mutable latest branch.